Vasopressin(Argipressin)

REVIEW BY (MAY 2024)

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Key Points

- Used in vasoplegic shock and in the optimisation of the brainstem dead patients for organ donor and transplantation. (Unlicensed use).
- Different doses and concentrations used for each indication.
- Give via central line (or large peripheral vein with close monitoring if central not available)
- First line in the context of organ donation and used in combination with other vasopressors normally noradrenaline in vasoplegic shock.
- Use of cardiac output monitor such as PICCO® advised to guide management in vasoplegic shock.

Background

Vasopressin (also called Argipressin) is Anti Diuretic Hormone (ADH) and improves blood pressure by constriction of vascular smooth muscle. The use of vasopressin in septic shock is suggested by the Surviving Sepsis Guidelines in cases where the use of noradrenaline is not achieving the target MAP, or as a noradrenaline sparing agent when noradrenaline doses are high. Donor optimisation guidance from NHS Blood and Transplant advise its use to manage some of the pathophysiological features of brain-stem death.

Adverse Effects

Adverse Effects & Monitoring: General monitoring is as for any patient requiring vasopressors.

Peripheries: Peripheral ischaemia - monitor closely for any signs of reduced peripheral perfusion.
CVS: Myocardial ischaemia, infarction and cardiac arrest, arrhythmias, bradycardia.
GI: Gastrointestinal ischaemia, nausea, vomiting and abdominal cramps.
Renal: Fluid retention.
Other: Local or systemic allergic reactions including anaphylaxis, headache, tremor, sweating, vertigo, bronchospasm.

Administration

- Administer via central line as vasopressin has a low pH and may cause venous irritation and tissue damage in cases of extravasation.
- If central access not available then administer via a large peripheral vein monitoring insertion site for signs of phlebitis. Resite cannula if this occurs.
- The half-life of vasopressin is around 10-20 minutes so double pumping is not required when renewing the syringe.

Vasoplegic Shock Dose and Concentration.

- In vasoplegic shock consider use if patient not achieving MAP with 20mL/hour 8mg in 50mL noradrenaline or greater than 0.5micrograms/kg/min.
- Dilute one ampoule of **20units to 50mL with glucose 5%** to produce a solution of 20 units in 50mL (0.4units per mL)
- It can also be diluted with Sodium Chloride 0.9% if necessary.
- A low dose vasopressin infusion of 0.03 units/minute (4.5mL/hour of the 0.4units/mL solution) is recommended by the Surviving Sepsis Guidelines. The dose may be **started at 0.01 units/minute (1.5mL/hour) and titrated to 0.04 units/ minute (6mL/hour).**
- ***Discontinuation****:* Multiple studies describe clinically significant hypotension when vasopressin is discontinued prior to norepinephrine. When discontinuing therapy, consider slowly tapering by 0.01 units/minute (1.5mL/hr) every 30 to 60 minutes to reduce the risk of hypotension.

Donation after Brainstem Death (DBD) Donor Optimisation Dose and Concentration.

- **20 units in 40mls with 5% Glucose** (0.5units/mL).
- First line vasopressor in this situation.
- Commence vasopressin **0.5 – 4 units/hour which is 1-8mL/hr** where vasopressor required. May need higher doses at Consultant discretion. Wean or stop other catecholamine pressors as able.
- Aim MAP 60 – 80mmHg or as advised by the specialist nurse for organ donation (SNOD) or critical care Consultant.

Compatibility

Compatible infusion (it is assumed that medicines meet close to the vascular access device.): Adrenaline, amiodarone (glucose 5%), caspofungin (NaCl 0.9%), ciprofloxacin, cisatracurium, diltiazem, dobutamine, dopamine, fluconazole (glucose 5%), gentamicin, GTN, heparin, imipenem& cilastatin (NaCl 0.9%), insulin (NaCl 0.9%), lidocaine, linezolid, meropenem, metronidazole, milrinone, noradrenaline, pantoprazole, phenylephrine, piperacillin-tazobactam (Tazocin®), sodium bicarbonate, voriconazole.

Incompatible with furosemide and phenytoin.

For full details of compatibility please refer to Medusa or seek advice from Pharmacy.

References

1. Medusa – Injectable Medicines Guide. Version 3 Accessed 19th June 2020.
2. Electronic Medicines Compendium. Argipressin. https://www.medicines.org.uk/emc/product/4117/smpc (Accessed 5th April 2021).
3. UptoDate. Vasopressin. (Accessed 5th April 2021).
4. NHSBT. Donor optimisation - Guidance around selecting potential DBD donors. https://www.odt.nhs.uk/deceased-donation/best-practice-guidance/donor-optimisation/ (Accessed 8th April 2021).